Injecting pupil binary intensity map into the laboratory adaptive optics bench using phase-only LCoS-SLM device

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ABSTRACT

Laboratory Adaptive Optics (AO) benches are the backbone of experimental testing and verification of new AO designs and architectures. These testbeds are particularly important when exploring unknown factors in the development of new instruments and facilities like future extremely large telescope AO systems. One of the key elements for simulating the performance of such systems in a smaller scale laboratory environment is the ability of projecting the precise intensity mask on the pupil plane. This mask often has binary (black or transparent/reflective) patterns that mimic the secondary obscuration and spider design of the telescope. Precise implementation of such intensity masks on the bench is important since studying effects such as "island/petaling effect" are critically dependent on the correct down-scaling and precise representation of the spider structure. Using a physical mask for such an application is very challenging since manufacturing and installing such fine structure pieces are difficult and hard to use. It is also necessary to build a new physical mask for each telescope system or scale that is desired for the experiment. In this paper, we introduce two methods of using a phase only Liquid Crystal on Silicon Spatial Light Modulator (LCoS-SLM) device as an alternative option to precisely and relatively easily inject the custom intensity mask into an optical bench. By implementing these methods on the LOOPS bench AO facilities at LAM, we demonstrated that the contrast produced by both methods could be better than 2% (dark/bright ratio), which is sufficient for representing pupil obscuration in the majority of applications. We also show that by using one of these methods, it is possible to inject phase and binary intensity mask simultaneously which could greatly increase the versatility and ease of use of an experimental AO setup.

Keywords: adaptive optics, spatial light modulator, intensity modulation, focal-plane filtering

1. INTRODUCTION

AO has become a common subsystem of large ground-based observatories. This key technology compensates for the disturbing effects the atmosphere imposes on the resolving power of ground-based telescopes. Without AO, the resolving power of a giant ground-based telescope degrades to that of a small 30 cm backyard one. The only alternative to the AO technology is to send the telescope and its instrument suite to space, which is an order of magnitude more costly. More importantly, the current space launch technologies limit the aperture size of the largest space telescope to be smaller than the aperture of future generation of extremely large ground-based telescopes by a factor of 5. This translates to 5 times less diffraction-limited resolving power and 25 times less light-gathering power, significantly limiting the critical science cases. This makes ground-based astronomical AO the main focus and research priority for the future of astronomy.

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Laboratory AO benches are the backbone of research and development in the field. These setups provide control over many degrees of freedom which are not possible to control within observatory or on-sky systems (i.e. controlled/slow input turbulence, controlled light intensity and wavelength etc). Almost all AO developments and instruments have a laboratory prototype counterpart.[1–5] However, emulating AO system performance in a laboratory environment is a challenging task. Mimicking the effect of atmospheric turbulence on the wavefront; simulating the telescope structure and flexure; and also the environmental conditions of the dome are a few of these challenges.

One of the important features to include in laboratory AO setups is the ability of providing accurate pupil shape and structure. Currently, the main method for fabricating large diameter mirrors for astronomical telescopes is the "*segmented mirror*" (i.e. Keck, the Thirty-Meter Telescope (TMT), the Extremely Large Telescope (ELT) etc). In a segmented mirror, the typical shape of each cell is a hexagon, causing jagged features on the edge of the pupil of the telescope. Furthermore, all large ground-based telescopes utilise a secondary mirror in front of their main primary aperture. This mirror is supported with a mechanical structure commonly referred to as the "spider" (see Fig. 1).

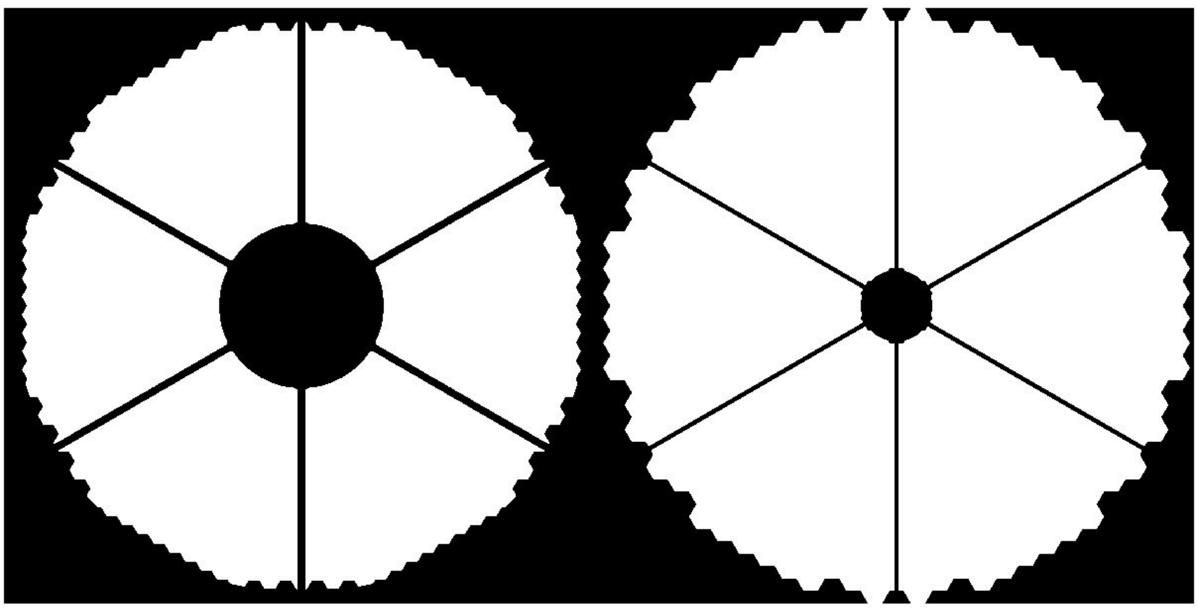

Figure 1. Pupil shape and spider shadow for ELT (Left) and TMT (right). Note the spider structure shadow on the primary and the jagged features on the edge of the pupil. Note that the scale in this figure is arbitrary. ELT and TMT primary mirrors will be 39 and 30 meters in diameter, respectively.

Building a precise, small-scale model of the intensity pattern created by the shadow of the spider structure and the exact shape of the telescope mirror within the laboratory setup is becoming increasingly important for gaining a deeper understanding of AO system performance. With this knowledge, we can better study and understand important challenges such as the low-wind effect,[6] the petaling effect[7] and the effect of partially illuminated pixels/sub-apertures on AO performance.[8,9] This small scale pupil intensity pattern modelling is a particularly challenging task as the ratio of the thinnest element of the spider structure to the pupil diameter can be of the order of 1%.

The current method for injecting pupil intensity pattern into the laboratory bench is the fabrication of an intensity mask[10] and placement of such a structure on a pupil-conjugated plane. Considering the scale of a typical laboratory bench and beam size, this can easily translate to sub-millimeter elements in the pupil mask structure which are hard to fabricate, expensive and difficult to work with. Additionally, the precise control of the position and clocking of the mechanical mask on a laboratory setup for emulating pupil rotation or wobbling is also a challenge with this approach. Finally, with the mechanical fabrication approach, each mask is specifically built for a specific intensity pattern. This means that a new mask is necessary to model for each different telescope.

In this paper we propose a new solution to this challenge based on SLM technology. This solution does not need any mechanical fabrication and provides ultimate flexibility to shift or rotate or change the intensity pattern.

2. METHOD

2.1 Spatial Light Modulator and phase modulation

There are different types of SLM devices with different physical properties. The most common and commercially available devices are the phase-only LCoS-SLMs.[11] The refraction index of some special materials such as liquid crystal can be changed by applying voltage. By utilizing such a material, LCoS-SLMs devices are able to control the phase of the reflected wavefront. Originally, SLM technology has been used in optical communication and optical projection applications,[12] but is being used more commonly for phase modulation on AO laboratory benches.[13–16] SLMs can provide similar functionality as deformable mirrors (DMs) in such setups. The main differences between the two are:

- Typically, SLM devices that have many more controllable phase units than DMs are significantly cheaper.
- Since deformable mirrors are based on reflection, the y are achromatic. The performance of SLM devices is a function of wavelength and therefore they are typically used in monochromatic setups.
- The response time of commercial SLMs is on the order of 10 ms, which is significantly slower than typical deformable mirrors.
- The stroke of each controllable phase unit (i.e. actuator, liquid crystal pixel) of SLM devices is significantly smaller than deformable mirrors, however there are methods to emulate high dynamic range phase patterns with SLMs by using phase wrapping.

Although SLM technology is not yet suitable for on-sky applications, it is an ideal choice to serve as a high-resolution, affordable deformable mirror on monochromatic laboratory benches. Utilising two SLMs conjugated to the pupil and focal plane, the LAM/ONERA Pyramid Sensor (LOOPS) optical bench[15,17,18] (see Fig. 2) in the LAM provide a comprehensive testbed for AO studies.

The focal plane-conjugated SLM makes it possible to extensively study Fourier-based wavefront sensing techniques such as generalised forms of Pyramid WaveFront Sensors (PWFS)[13,14] while the pupil-conjugated SLM is used for phase modulation purposes such as injecting atmospheric or custom phase turbulence into the optical path.[19] This architecture along with the utilisation of parallel Shack-Hartmann WaveFront Sensor (SHWFS) and advanced detectors provides the LOOPS bench with the flexibility for a wide range of AO research and studies. In collaboration with LAM, we designed and demonstrated two experimental approaches to use the pupil-conjugated SLM for intensity pattern injection, adding new abilities to the LOOPS bench.

2.2 Approach A: Cross polarization

LCoS-SLM device modulates the phase, only across a certain polarization vector. The perpendicular polarization component of the beam would remain untouched and should be filtered from the system. In the cross polarization approach, the device is placed between two perpendicular linear polarisers with 45° angle between the polarization vector of the SLM and filters. The fist polariser would be placed before the SLM device, turned 45° respective to the working vector of the SLM. The second polariser is placed after the SLM, turned 45° respective to the working vector of the SLM and 90° respective to the first polariser filter. Without modulation, no light would pass through the system. However, applying voltage to the SLM pixels changes the polarization between the two perpendicular filters, hence allowing some of the light to pass through. The schematic of this method is shown in Fig. 3.

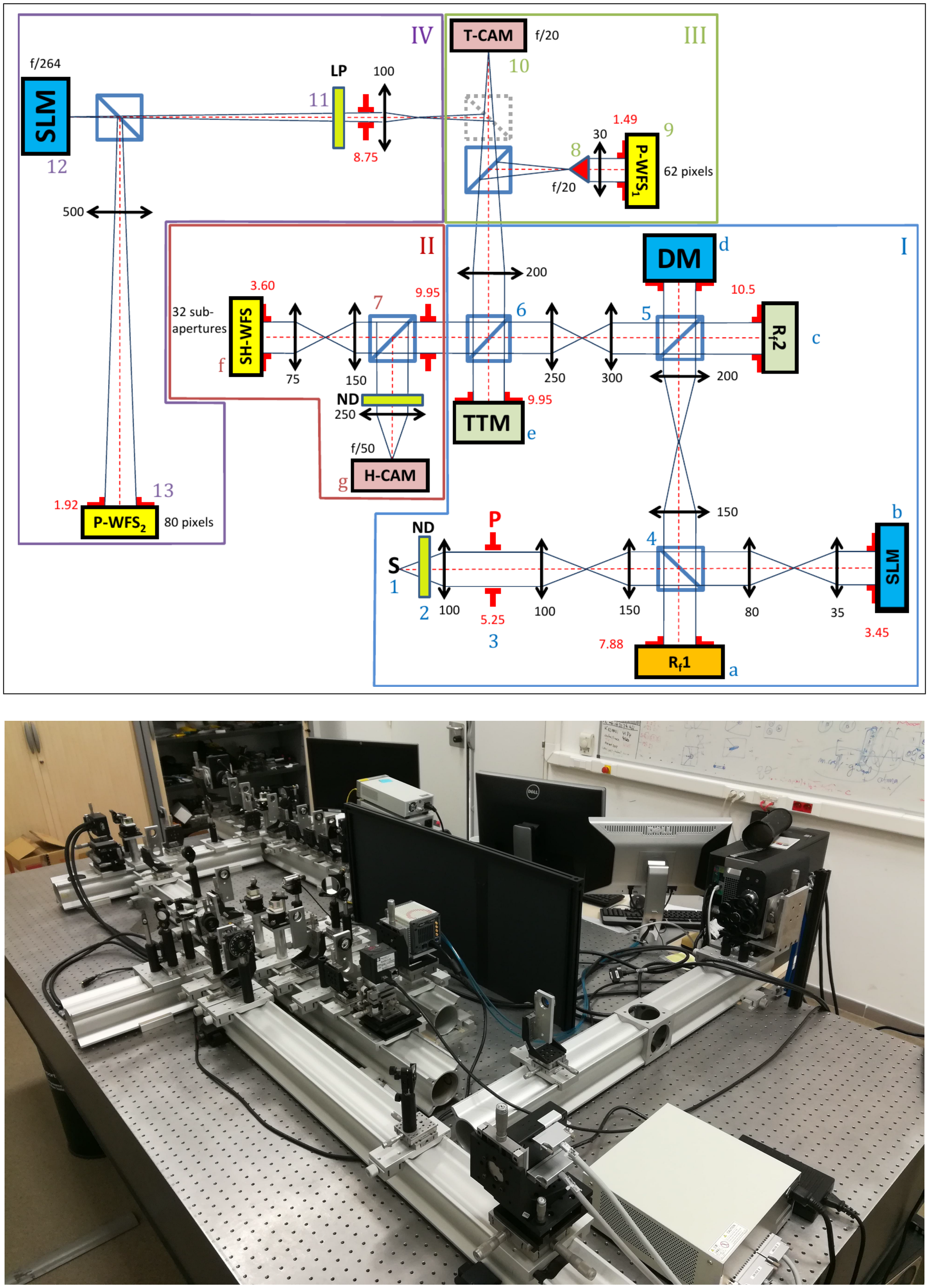

Figure 2. The LOOPS: Schematic design and laboratory setup.

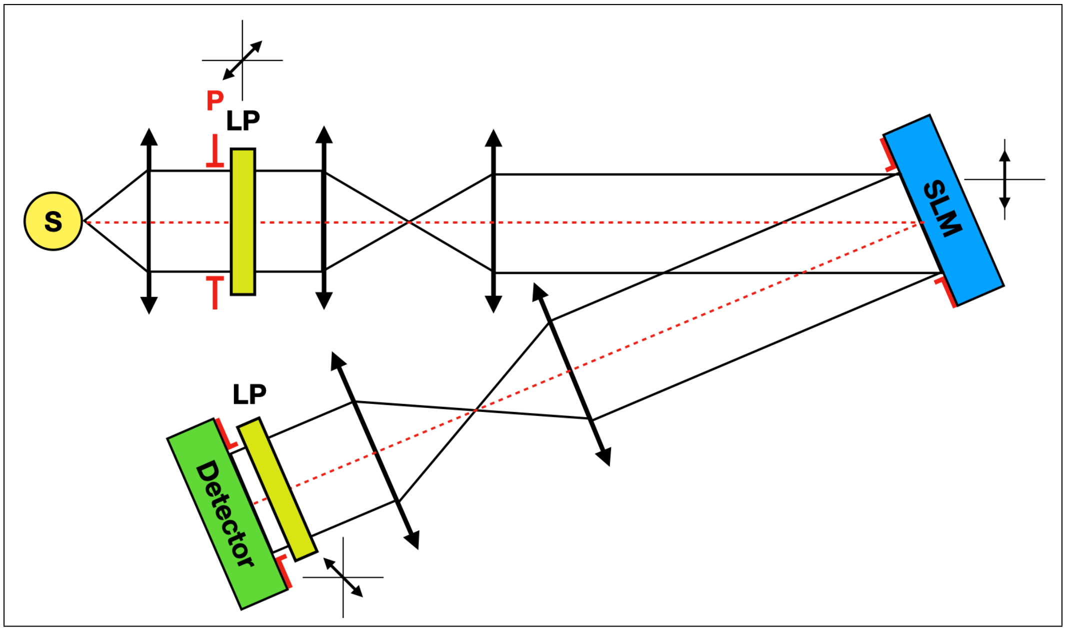

Figure 3. The schematic of the cross polarization approach.

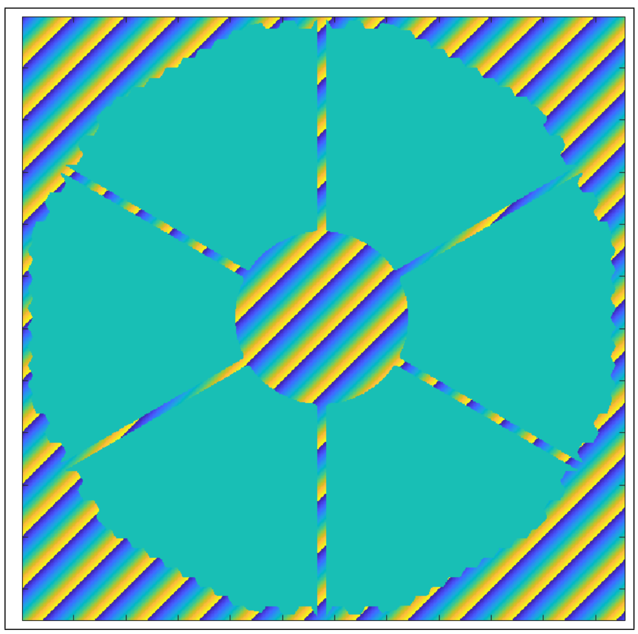

Figure 4. An example of pupil SLM phase modulation pattern for ELT pupil and spider structure. The light from the shadow region is diverted to a large separation of the optical axis using a phase wrapped high angle tilt-tip pattern. The illuminated region of the pupil (the flat green region) can be used to inject a custom phase pattern into the system.

2.3 Approach B: Focal-plane filtering

In the focal plane filtering approach, the light in the shadow region is diverted to a certain point in the focal plane by applying an intense tilt/tip pattern (see Fig. 4). The off-axis illumination pattern is then filtered in the focal plane using a spatial filter such as an iris. This approach also filters out the non-modulated and non-polarised portion of the light resulting from the realistic SLM fill factor. SLM pixels are not ideal and there are typically 4-10% of the surface area that are not optically active. The reflected light from this region reduces the contrast and increases the noise of the experiment results. To filter out this residual, it is possible to add a tilt to the desired phase modulation pattern and filter the non-modulated part using an off-axis iris. It is possible to add intensity pattern injection by utilising the same concept, diverting shadow light to the opposite direction of the focal plane and filter it the same way. A simple schematic of such a layout is presented in Fig. 5.

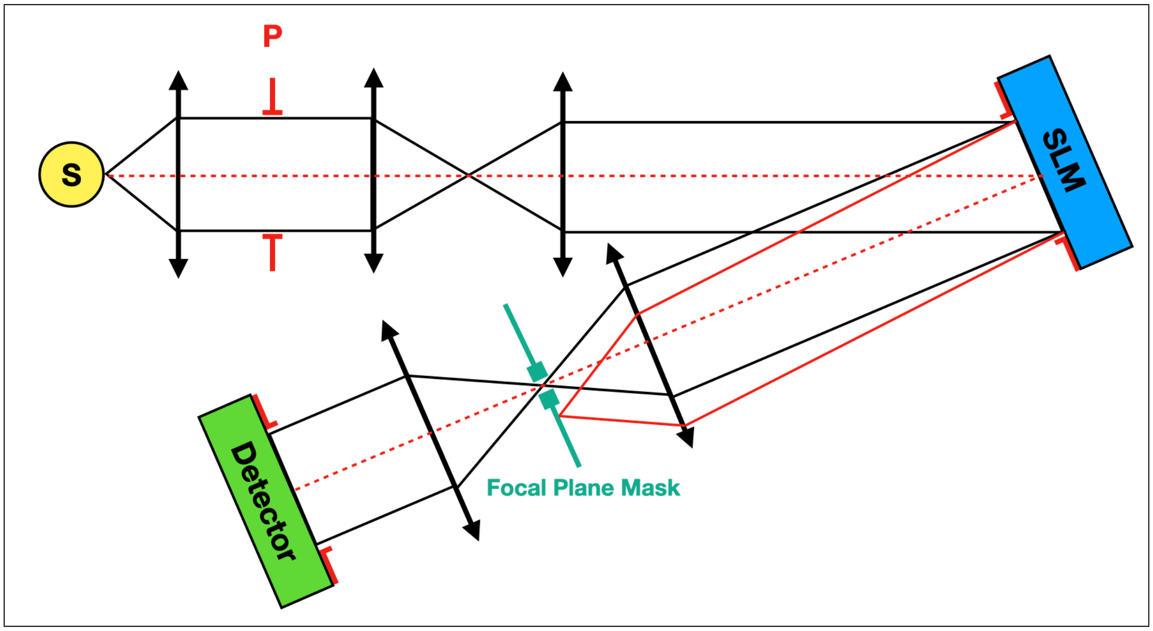

Figure 5. Schematic of the focal plane filtering approach. The red beam contains the light from the shadow area which is tilted and filtered in the focal plane.

3. RESULTS AND METHOD COMPARISON

We successfully implemented both methods on the LOOPS bench. Both methods provide similar contrast (the ratio of the normalised flux in the shadow to the illuminated region) sufficient for the majority of AO applications. However the other considerations such as complexity and the ability of simultaneous phase and intensity modulation make them distinct.

The result for the cross polarization method is shown in Fig. 6. By fine tuning the phase shift of SLM pixels and the angle between linear polarisers, the contrast of 2% is achieved. This method is easier to implement but two focal plane conjugated SLMs would be necessary for simultaneous phase and intensity control. It should be noted that producing gray scale illuminated patterns (i.e. simulating atmospheric scintillation effect) is not possible with this approach as any shift in intensity would be accompanied with an undesired phase modulation.

Using the focal plane filtering approach, a better contrast of 0.5% can be achieved. However, this method requires more accurate fine tuning. The most sensitive alignment aspect is to correctly account for the tilt/tip of the shadow region and the size of the off-axis pupil in a way to filter out undesired light and simultaneously completely pass through the desired Point Spread Function (PSF) and its wings. This is more difficult especially when considering higher-order diffraction spots caused by phase wrapping. An example of such spots can be seen in the top left panel of Fig. 8. The amount of tilt/tip for the shadow region should be carefully chosen in the way that the off axis spot be significantly far from the main PSF wings and at the same time, there be no high-order diffraction spot on the top of the main PSF. The major benefit of this method is the ability of using the same SLM for simultaneous phase and intensity modulation. The result for this approach can be seen in Fig. 7.

Since simultaneous phase modulation is possible in this, we also closed the loop while injecting ELT's pupil structure and spider pattern. The simulated and on-bench PSF can be seen in the top panel of Fig. 8, as well as high-resolution images of the pupil intensity pattern for ELT and TMT in the bottom panel.

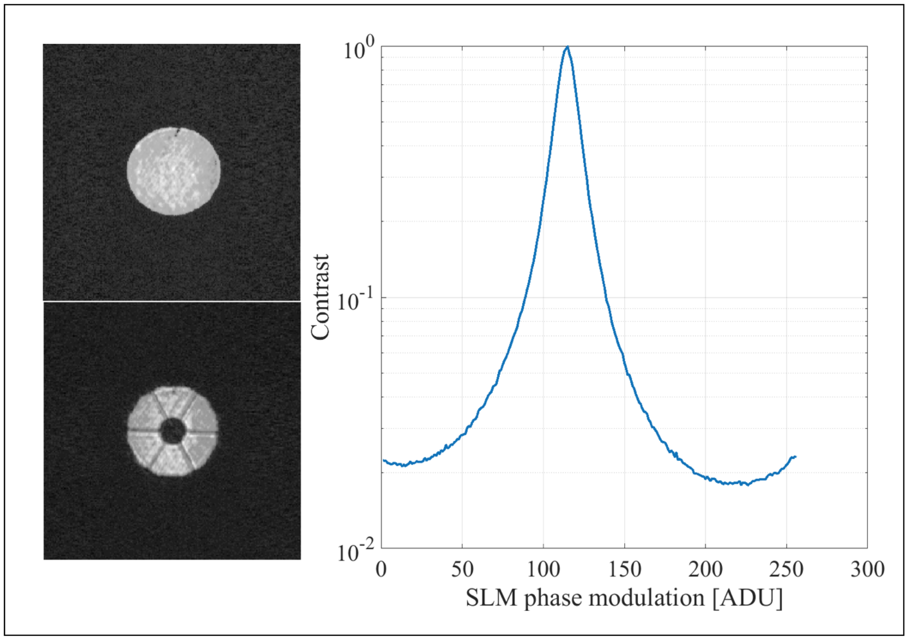

Figure 6. The result of the cross polarization approach. Left: the pupil intensity before and after injecting the intensity pattern. Right: The contrast per SLM phase input. The contrast of 2% is achieved through this method. The E-ELT pupil shape and spider shadow (see Fig. 1) is used as the intensity pattern.

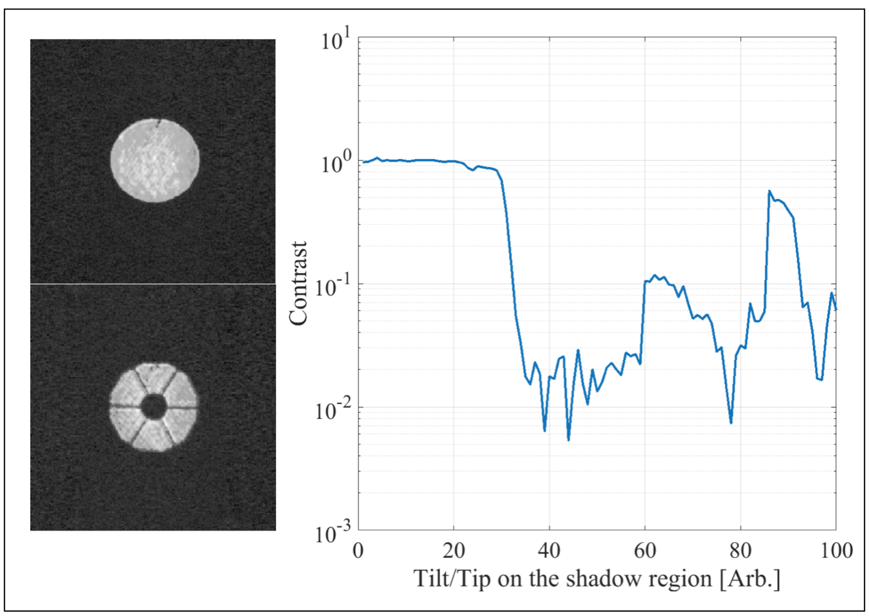

Figure 7. The result of the focal plane filtering approach. Left: The pupil intensity before and after injecting the intensity pattern. Right: The contrast per input tilt for shadow region. In the optimum configuration, contrast of 0.5% is achieved. The ELT pupil shape and spider shadow (see Fig. 1) is used as the intensity pattern. Note that the contrast curve is irregular in comparison with the cross polarisers approach. This is due to higher order diffraction pattern caused by the SLM phase wrapping. This particularly makes the fine tuning focal plane approach relatively more challenging.

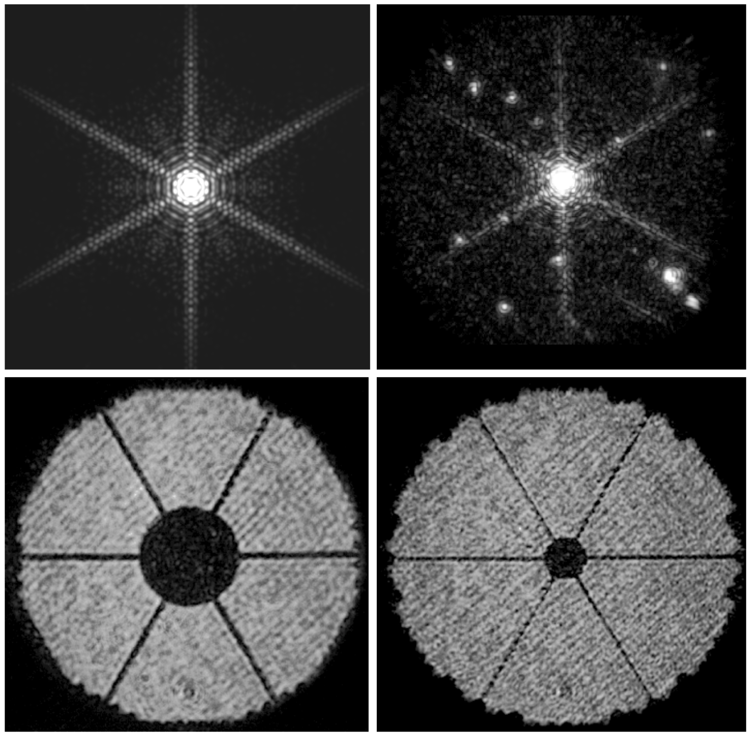

Figure 8. *The top panel:* Simulation and close loop demonstration of the PSF shape for ELT pupil structure. The close similarities between the two PSF demonstrates the success of the intensity pattern injection using the SLM device. The vignetting on the top tight image is caused by the focal plane filter/iris. Note the few fainter spots at 4 and 10 o'clock of the main PSF. They are ghosts of high-order diffraction spots caused by the SLM phase-wraped tilt/tip pattern. *The bottom panel:* The high resolution pupil images demonstrated on the LOOPS bench for the intensity structure of ELT (left) and TMT (right) pupils. The faint diagonally striped pattern is caused by the internal diffraction of the beam splitters and is not related to SLM or intensity pattern injection.

4. SUMMARY AND DISCUSSION

We successfully demonstrated two different approaches to inject high-precision intensity patterns into the LOOPS AO bench. These methods are based on phase-only LCoS SLM technology. This technology has some limitations such as wavelength dependent modulation coefficient, however the significantly higher resolution and affordability make it a very desirable substitute of deformable mirrors for a wide range of experiments. The ability of emulating the precise pupil shape and the effect of spider shadow on the performance of the AO system and PSF shape is essential for advanced AO instrument laboratory setups. Study of important challenges such as the effect of partially illuminated pixels on the performance of PWFS and PFS shaping for high-contrast imaging are strongly dependent on such an ability.

In comparison, the focal plane filtering method has two major benefits: the ability of simultaneous phase and intensity control using only one SLM device and the removal of non-modulated light leaked by the not-ideal filing factor of SLM pixels. However, tighter alignment tolerances make it relatively more difficult to implement. This is because two simultaneous effects should be considered in the alignment of the focal plane iris in this method. On the one hand the iris should be as wide as possible to not to filter the wings of the desired PSF, which is particularly difficult in intense high-tilt phase patterns. On the other hand, the size of the pupil should be smaller to not let any of the higher order diffraction pattern caused by SLM phase wrapping pass through the focal plane or superimposed by the desired PSF. This makes this method more suitable for the tilt/tip removed or closed-loop residual phase scenarios.

The cross-polarization method on the other hand is more straight forward to setup, although in our experience the resulted contrast is slightly less than the focal plane filtering approach. It should be noted that both approaches provide the ability of injecting binary intensity patterns and are not suitable for emulating gray scale intensity phenomena like atmospheric scintillation.

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